Kendall Thomas

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Education

- **University of North Carolina at Chapel Hill** — **Est. May 2027**, Chapel Hill, NC
  - Ph.D. Statistics and Operations Research (Advisor: Jan Hannig)
  - Dissertation Title: Statistical Frontiers in Elite Athlete Performance

- **University of North Carolina at Chapel Hill** — **Dec. 2025**, Chapel Hill, NC
  - M.S. Statistics, Analytics, and Data Science

- **Davidson College** — **May 2019**, Davidson, NC
  - B.S. Mathematics, B.S. Computer Science
  - Magna Cum Laude

Professional Experience

- **Graduate Teaching Fellow** — **Jan. 2026 – Present**
  - *University of North Carolina* — **May 2024 – July 2025**
  - *Instructor of Record: STOR 320, STOR 455* — Chapel Hill, NC
- **Graduate Research Assistant** — **Aug. 2025 – Dec. 2025**
  - *University of North Carolina* — Chapel Hill, NC
- **Graduate Teaching Assistant / Lab Instructor** — **Aug. 2022 – May 2024**
  - *University of North Carolina* — Chapel Hill, NC
- **Software Engineer (promoted to Software Engineer II)** — **Aug. 2019 – Apr. 2022**
  - *Microsoft, Azure Data* — Redmond, WA
- **Research Assistant** — **Aug. 2016 – May 2019**
  - *Dr. Tim Chartier, Davidson College* — Davidson, NC
- **Assistant Analyst** — **Jan. 2018 – Dec. 2019**
  - *John Tobias, ESPN and FOX Sports* — Remote
- **Advanced Analytics and Machine Learning Intern** — **May 2018 – Aug. 2018**
  - *Nike* — Beaverton, OR
- **Research Assistant** — **Feb. 2017 – Nov. 2017**
  - *Athlete Intelligence* — Seattle, WA / Remote

Publications

1) Thomas, K. L., & Hannig, J. (2025). Movement dynamics in elite female soccer athletes: The Quantile Cube Approach. *Journal of Quantitative Analysis in Sports.* Advance online publication. https://doi.org/10.1515/jqas-2025-0014.[1]

Teaching

Curriculum Developed

- "Introduction to Data Science in Python"
    1) Led a curriculum redesign introducing a Python-based section of STOR 320 *Methods and Models of Data Science* (first offered Summer 2024; renamed from Introduction to Data Science in Fall 2025) alongside the existing R-based version.
    2) Developed new lectures and Jupyter-based lab modules emphasizing data acquisition, cleaning, visualization, modeling, and communication in Python.
    3) The Python section is now offered each semester, providing students an alternative track aligned with modern data science tools and reproducible workflows.

Courses Taught (Instructor of Record)

- **Summer Session II 2026** — 7 students

  1) STOR 455 (Methods of Data Analysis)
- **Spring 2026** — 42 students

  2) STOR 320 (Methods and Models of Data Science)
- **Summer Session II 2025** — 15 students

  3) STOR 455 (Methods of Data Analysis)
- **Spring 2025** — 94 students

  4) STOR 320 (Introduction to Data Science)
- **Fall 2024** — 101 students

  5) STOR 455 (Methods of Data Analysis)
- **Summer Session II 2024** — 22 students

  6) STOR 320 (Introduction to Data Science)

Awards and Honors

Academic

- **Druscilla French Graduate Student Excellence Fund Award** — **Aug. 2025 – May 2026**
    - *University of North Carolina at Chapel Hill*
- **Walter Deemer Excellence in Teaching Award** — **Dec. 2024**
    - *University of North Carolina at Chapel Hill Department of Statistics and Operations Research*

- **Excellence in Teaching Assistance and Instruction Award** **Dec. 2023**
    - *University of North Carolina at Chapel Hill Department of Statistics and Operations Research*

Conference & Research Competitions

- **First Place Poster Award**[1] **Sept. 2025**
    - *New England Symposium on Statistics in Sports (NESSIS)* Boston, MA
- **First Place Poster Award**[1] **Nov. 2024**
    - *Carnegie Mellon Sports Analytics Conference* Pittsburgh, PA
- **Hackathon Finalist (Professional Division)** **Mar. 2020**
    - *MIT Sloan Sports Analytics Conference* Boston, MA
- **Grace Hopper Celebration Research Scholar** **Oct. 2017**
    - *Computing Research Association – Women* Orlando, FL

Industry & Technical Hackathons

- **Winner - Greatest Business Value** **Fall 2021**
    - *Microsoft Azure Supportability Hackathon*
- **Winner – Most Impactful & Greatest Business Value** **Spring 2021**
    - *Microsoft Azure Supportability Hackathon*
- **Winner – Best Hack (Audience Vote)** **Fall 2019**
    - *Microsoft Azure Data Hackathon*

Leadership, Service, & Outreach

- **Graduate Student Liaison** **Jan. 2024 – Present**
    - *University of North Carolina at Chapel Hill Department of Statistics and Operations Research*
- **Graduate Student Mentor** **Aug. 2023 – Present**
    - *University of North Carolina at Chapel Hill Department of Statistics and Operations Research*
- **Sports Analytics Intelligence Lab (SAIL) - Principal** **May 2025 – Present**
- **Sports Analytics Intelligence Lab (SAIL) - Graduate Student Lead** **Jan. 2023 – Present**
    - *University of North Carolina at Chapel Hill Department of Statistics and Operations Research*

- **Varsity Athlete NCAA Division I Women's Soccer Team** **July 2015 – May 2019**
    - *Davidson College*

[1] Related Paper, Presentations, and Awards

- **Analyst & Team Manager NCAA Division I Men's Basketball Team** **Jan. 2018 – May 2019**
  - *Davidson College*
- **'Cats Stats Lead Analyst** **Jan. 2016 – May 2019**
  - *Davidson College*
- **Females in Computer Science and Information Technology President** **Aug. 2017 – May 2019**
  - *Davidson College*

Presentations

Invited Talks

- **"Analyzing Movement Dynamics in Elite Female Soccer Athletes"[1]** **Apr. 2026**
  - Related Research Authors: Thomas, K. L., & Hannig, J. Mansfield, CT
  - *Connecticut Sports Analytics Symposium*
- **"Analyzing Movement Dynamics in Elite Female Soccer Athletes"[1]** **Jan. 2026**
  - Related Research Authors: Thomas, K. L., & Hannig, J. Houston, TX
  - *American Soccer Insights Summit*
- **"Beyond a Bit Fit"** **Apr. 2019**
  - Related Research Authors: Thomas, K. L. & Chartier, T. Greenville, SC
  - *Carolina Sports Analytics Meeting*

Conference Presentations

- **"Does Age Shape How Athletes Move? Extending the Quantile Cube Framework to Detect Age Effects in Movement Distribution Profiles"** **Scheduled Aug. 2026** Boston, MA
  - Related Research Authors: Thomas, K. L., & Hannig, J.
  - Topic-Contributed Session: "Statistical Methods for Estimating Player Performance Across Ages and Leagues"
  - *Joint Statistical Meetings (JSM)*
- **"Analyzing Movement Dynamics in Elite Female Soccer Athletes"[1]** **Apr. 2026**
  - Related Research Authors: Thomas, K. L., & Hannig, J. Virtual
  - *ASA Virtual Sports Analytics Conference*
- **"Analyzing Movement Dynamics in Elite Female Soccer Athletes"[1]** **Nov. 2024**
  - Related Research Authors: Thomas, K. L., & Hannig, J. Virtual
  - *Midwest Sports Analytics Meeting*

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[1] Related Paper, Presentations, and Awards

- **"Beyond a Bit Fit"** **Nov. 2019**
  - Related Research Authors: Thomas, K. L. & Chartier, T. Pella, IA
  - *Midwest Sports Analytics Meeting*

Conference Posters

- **"Analyzing Movement Dynamics in Elite Female Soccer Athletes"[1]** **Feb. 2026**
  - Thomas, K. L., & Hannig, J. Chapel Hill, NC
  - *CoreAI UNC Research Showcase*
- **"Analyzing Movement Dynamics in Elite Female Soccer Athletes"[1]** **Sept. 2025**
  - Thomas, K. L., & Hannig, J. Boston, MA
  - *New England Symposium on Statistics in Sports*
- **"Analyzing Movement Dynamics in Elite Female Soccer Athletes"[1]** **Nov. 2024**
  - Thomas, K. L., & Hannig, J. Pittsburgh, PA
  - *Carnegie Mellon Sports Analytics Conference*
- **"Professional vs. Elite Female Soccer Athletes: Physical and Technical Performance Characteristics"** **Jul. 2024** Baltimore, MD
  - Cantú, E. I., Moore, S. R., Thomas, K. L. & Smith-Ryan, A. E., FNSCA.
  - *National Strength and Conditioning Association National Conference*

Professional Memberships

- American Statistical Association (ASA)
- Association for Women in Mathematics (AWM)
- Phi Beta Kappa Society
- Omicron Delta Kappa: The National Leadership Honor Society

Technical Skills

Programming Languages

- Advanced: Python, R
- Intermediate: Java, HTML, CSS, SQL, Kusto
- Basic: C++, C#, C, JavaScript, p5.js, d3.js

Software and Tools

- Advanced: LaTeX, R Studio, Microsoft Office, Tableau, Jupyter Notebooks
- Intermediate: SQL Server Management Studio, Azure Data Studio
- Basic: Mathematica, ArcGIS, MATLAB

Libraries and Frameworks

- Data Manipulation & Analysis: pandas, NumPy, SciPy
- Data Visualization: ggplot, matplotlib, Plotly, seaborn
- Interactive Applications: Shiny (R and Python)
- Machine Learning / Data Science: scikit-learn, TensorFlow, PyTorch
- Web Scraping & Automation: BeautifulSoup, requests, Selenium
- Version Control / Workflow: Git / GitHub